

# Pole detection at the carrier: the beyond-endoscopy dichotomy for transported representation data

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## Abstract

Langlands' beyond-endoscopy principle detects functorial origin through poles:  $L(s, \pi, r)$  has one at  $s = 1$  exactly when  $r$  contains the trivial representation. This note compiles that detection for transported data  $\rho = \oplus_i \text{Sym}^{r_i}$  of the level-one seed at the mechanism's package level. Unconditionally, a trivial constituent splits a  $\zeta$ -factor off the datum bank, explicit in the  $L$ -readout on the joint half-plane. The detection theorem: *a datum containing the trivial representation admits no strict theta reflection* — from the strict joint identity the mechanism yields an entire completed transform, while the  $\zeta$ -constituent forces the simple pole at  $s = 1$ ; analytic continuation across  $\{\text{Re } s > 1\}$  and a ray limit against the residue collide the two, unless the completed erased factor vanishes at the edge, which is the named classical input. Contrapositive: a datum satisfying the strict, mass-free reflection has no trivial constituent. The masses of the weak reflection pair are the carrier avatar of the pole:  $\zeta$ -containing objects reflect only with mass terms.

## 1 The $\zeta$ -factorization

Fix the level-one eigenform seed with Hecke data, and a representation datum  $\rho = [r_1, \dots, r_m]$  with  $0 \in \rho$ . Both statements are unconditional:

**Lemma 1** (`repBank_zeta_factor`, `LSeries_repBank_zeta_factor`; machine-checked). *The datum bank splits as*

$$c_\rho = \zeta \star c_{\rho \setminus \{0\}},$$

*the rank-zero clock being the unit system (`symrBank_zero_eq_zeta`); on the joint half-plane*

$$L(\rho, s) = \zeta(s) \cdot L(\rho \setminus \{0\}, s),$$

*with  $\zeta$  the Riemann zeta function of `Mathlib` (`LSeries_zeta_eq_riemannZeta`).*

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\*Machine-checked companion: `RequestProject/PoleDetection.lean`, 4 named declarations, each at axiom footprint `{propext, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

## 2 The detection theorem

**Theorem 2** (`trivial_constituent_detection`; machine-checked). *Let  $0 \in \rho$ . Suppose, at some general Deligne chart  $(\delta s, \mu s)$  with nonnegative shifts:*

1. *the strict joint theta identity of the prescribed  $\rho$ -readouts (the input of the mechanism, with zero masses);*
2. *a package  $Q$  for the erased datum  $\rho \setminus \{0\}$  at the same chart (entire  $\Lambda_Q$ , functional equation, chart identification); and*
3. *edge nonvanishing:  $\Lambda_Q(1) \neq 0$ .*

*Then **False**.*

The proof is the collision of two compiled facts. From (1), `rep_general_package_of_theta` yields an *entire*  $\Lambda_P$  with  $\Lambda_P(s) = G(s) L(\rho, s)$  on the joint half-plane; by Lemma 1 and the chart of  $Q$ ,

$$\Lambda_P(s) = \zeta(s) \cdot \Lambda_Q(s) \quad (\text{Re } s \text{ large}),$$

and both sides are analytic on the connected open half-plane  $\{\text{Re } s > 1\}$ , so the identity persists there (`AnalyticOnNhd.eqOn_of_preconnected_of_eventuallyEq`). Along the real ray  $s = 1+t$ ,  $t \rightarrow 0^+$ :

$$(s-1) \Lambda_P(s) \longrightarrow 0 \quad \text{while} \quad (s-1) \zeta(s) \Lambda_Q(s) \longrightarrow \Lambda_Q(1)$$

(Mathlib’s `riemannZeta.residue_one` and continuity of the entire factors), whence  $\Lambda_Q(1) = 0$ , contradicting (3).

**Corollary 3** (contrapositive register). *A representation datum of the seed satisfying the strict theta reflection at an admissible chart — together with an erased-datum package nonvanishing at the edge — contains no trivial constituent. Equivalently: the trivial constituent is detected by the reflection type —  $\zeta$ -containing objects reflect only with mass terms, never strictly. The masses  $f_0, g_0$  of the weak reflection pair are the carrier-side avatar of the pole.*

## 3 Register

**Remark 4** (register). **Proved (machine-checked)**. *The  $\zeta$ -factorization of the datum bank and its  $L$ -readout; the threshold bound; the detection theorem exactly as stated, with its three inputs typed in the signature.*

**Consumed from the corpus.** *The mechanism at the general chart (`mechanismG` via `rep_general_package_of_theta`), datum self-duality, the compiled bank calculus. From Mathlib: the residue of  $\zeta$  at  $s = 1$ , differentiability of  $\zeta$  off  $s = 1$ , analytic-continuation uniqueness.*

**Cited, not formalized.** *The classical discharge of input (3) — edge nonvanishing of the completed erased factor — is the Jacquet–Shalika class of nonvanishing theorems [2]; for symmetric-power factors of low rank it is known unconditionally through the Kim–Shahidi lifts [3]. Input (2) is the mechanism’s own output given the erased datum’s identity.*

**Not claimed.** *No trace formula, no orbital integrals, no general automorphic  $\pi$ : the scope is the transported data of the level-one  $\text{GL}(2)$  seed, and “beyond endoscopy” names the detection principle compiled here — pole  $\leftrightarrow$  trivial constituent  $\leftrightarrow$  reflection type — not Langlands’ full program [1].*

**Falsifier.** *Exhibiting a datum with a trivial constituent that satisfies the strict identity, together with a nonvanishing erased package, would make the compiled theorem produce **False** — an inconsistency of the corpus with Mathlib. Classically, the constant term of the theta function of any  $\zeta$ -containing object is nonzero, which is why no such datum exists; the theorem is the compiled form of exactly that obstruction.*

## References

- [1] R. P. Langlands, *Beyond endoscopy*, in: Contributions to automorphic forms, geometry, and number theory, Johns Hopkins Univ. Press, 2004, 611–697.
- [2] H. Jacquet, J. A. Shalika, *A non-vanishing theorem for zeta functions of  $GL_n$* , Invent. Math. **38** (1976/77), 1–16.
- [3] H. Kim, F. Shahidi, *Functorial products for  $GL_2 \times GL_3$  and the symmetric cube for  $GL_2$* , Ann. of Math. (2) **155** (2002), 837–893.